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Privacy-preserving labeling-free occupancy counting sensor based on ToF camera and clustering

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Abstract

Occupancy detection systems are crucial for optimizing energy efficiency in smart cities and buildings but often face privacy and data dependency challenges. YOLO (you only look once), a widely used real-time detection framework, relies on identifiable image data and labeled datasets. This study proposes a privacy-preserving, labeling-free occupancy sensor using a time-of-flight (ToF) camera, and a clustering algorithm. Positioned above doorways, the ToF camera captures depth data that inherently protect privacy by avoiding identifiable information. Using the mean shift clustering algorithm, it performs real-time detection and tracking without labeled data, generating bounding boxes for movement analysis. Unlike traditional ToF-based or unsupervised methods, the proposed system adapts dynamically to varying occupant behaviors and environmental conditions for robust real-time detection. Experimental results show that the proposed method achieves over 90% accuracy in standard single-entry and exit scenarios. By addressing existing limitations, it offers a data-efficient, privacy-sensitive solution for building digital twins in energy optimization and resource management.

KEYWORDS

clustering algorithm, occupancy detection, privacy-preserving technology, time-of-flight (ToF) camera, unsupervised learning

1 | INTRODUCTION

The rapid advancement of smart building technologies, integral to the development of smart cities, has highlighted the need for sophisticated occupancy detection systems that optimize energy efficiency and improve spatial utilization [1]. These systems play a vital role in dynamically managing building operations, such as heating, ventilation, air conditioning (HVAC), and lighting, by responding to real-time occupancy data [2, 3]. By reducing unnecessary energy consumption, they

contribute to significant cost savings, lower carbon footprints, and enhanced occupant comfort. Furthermore, such technologies are pivotal for implementing building digital twins, enabling precise energy optimization and efficient resource management [4]. Real-time occupancy information can be directly utilized in building energy optimization simulations within digital twin models, enhancing their effectiveness and accuracy.

One of the most widely adopted frameworks for object detection and occupancy monitoring is YOLO (you only look once), known for its real-time processing capability

and high accuracy in detecting and classifying multiple objects within complex environments [5]. YOLO's capabilities have made it a popular choice in dynamic settings such as building energy efficiency [6]. However, YOLO and similar vision-based frameworks heavily rely on RGB image data, which captures detailed visual information, including facial features and other sensitive attributes. This reliance raises significant privacy concerns, making such systems unsuitable for environments where privacy preservation is paramount, such as homes, offices, and other facilities [7]. Moreover, the use of high-resolution RGB cameras often leads to increased storage and computational requirements, further limiting their scalability in resource-constrained environments.

In addition to privacy concerns, YOLO and other supervised learning-based methods depend on large volumes of labeled datasets for training, which are time-intensive and costly to produce [8, 9]. Fine-tuning YOLO for occupancy detection requires collecting diverse image datasets that reflect the target environment and occupant behaviors, a process that demands substantial human and time resources [10]. Additionally, this process must be repeated whenever the operational environment changes, such as when deploying the system in buildings with different architectural layouts or when occupant behaviors deviate from the original dataset [11]. Each new deployment introduces unique challenges, requiring additional labeled datasets and retraining of the model. This dependency not only increases deployment costs but also restricts the adaptability of YOLO-based systems to diverse application scenarios, particularly in environments with dynamic occupant behaviors. These limitations underscore the need for innovative occupancy detection systems that operate without reliance on labeled data while addressing privacy concerns [12].

Recent advancements in depth-sensing technologies, such as time-of-flight (ToF) cameras, and unsupervised learning techniques, such as clustering algorithms, offer promising solutions to the challenges of privacy-preserving and adaptive occupancy detection. ToF cameras capture depth data rather than visual details, inherently preserving privacy by avoiding the collection of personally identifiable information [13]. Compact, cost-effective, and capable of operating in diverse lighting conditions, ToF cameras are well-suited for real-world deployment, particularly in privacy-sensitive environments [14, 15]. While previous studies have employed ToF cameras for static tracking [16], activity recognition via sensor fusion [17], and integration with radar sensors for occupancy grids [18], these approaches largely relied on rule-based methods or static techniques, limiting their adaptability to dynamic environments and diverse occupant behaviors. To address these limitations,

machine learning techniques, such as those employed in YOLO, are crucial for enabling systems to generalize across varying conditions.

Complementing the privacy-preserving capabilities of ToF cameras, unsupervised learning techniques further address the challenges associated with data dependency. Unlike supervised methods, which require labeled datasets, unsupervised algorithms such as clustering can identify patterns and structures directly from raw data. These techniques eliminate the need for manual labeling, reducing costs and increasing scalability. Recent studies have applied unsupervised methods like principal component analysis (PCA) to masked face recognition [19, 20] but focus on feature-level classification rather than pixel-level clustering for occupancy detection. While previous studies have applied unsupervised learning for occupancy detection using sensor data, such as air quality or motion sensors [21–23], these approaches often lacked real-time detection capabilities or were confined to specific modeling scenarios. To address these limitations, YOLO-based systems have emerged as a solution for real-time processing; however, our goal is to combine the advantages of YOLO's real-time capabilities with the strengths of unsupervised learning, enabling adaptability and reduced dependency on labeled datasets.

This study introduces a privacy-preserving, labeling-free occupancy detection system that uniquely integrates the advantages of ToF cameras with unsupervised learning algorithms. Unlike prior ToF-based systems that often rely on rule-based approaches or static sensor fusion techniques, the proposed system dynamically adapts to diverse occupant behaviors and environmental conditions through the use of unsupervised clustering. Specifically, the system employs a ToF camera installed above doorways to capture depth data, ensuring that no personally identifiable information or interior details are recorded, thereby inherently complying with stringent privacy regulations. By utilizing the mean shift clustering algorithm, a robust unsupervised learning technique, the system enables real-time occupant detection and tracking without the need for labeled datasets, overcoming a key limitation of many unsupervised approaches that focus solely on static analysis or modeling. Additionally, bounding box generation and movement tracking mechanisms enhance the system's capability for accurate occupancy counting and analysis, addressing the critical requirements of dynamic and privacy-sensitive smart building applications.

The contributions of this research are threefold:

- **Privacy Preservation:** The system inherently preserves privacy by using ToF cameras, which prevent the collection of identifiable image data, ensuring compliance with privacy regulations.

- **Label-Free Detection:** The integration of unsupervised learning algorithms eliminates the need for labeled datasets, reducing time, cost, and complexity while enhancing adaptability.
- **Scalability and Efficiency:** The system achieves high accuracy in low-resource environments, offering real-time performance and robust adaptability to varying conditions.

The rest of this paper is structured as follows. Section 2 details the proposed methodology, focusing on the integration of ToF cameras and the mean shift clustering algorithm. Section 3 presents the experimental setup and results, evaluating the system's performance across various scenarios. Finally, Section 4 concludes the study and outlines directions for future research.

2 | PROPOSED METHODOLOGY

2.1 | ToF camera for privacy-preserving depth data collection

The proposed system utilizes ToF cameras to ensure privacy preservation in occupancy detection. Unlike RGB cameras, which capture detailed visual information including facial features and other identifiable attributes, ToF cameras produce depth data, where each pixel represents the distance to an object in the camera's field of view [13]. These data are inherently nonidentifiable, ensuring compliance with privacy regulations. Figure 1 illustrates the difference between the two approaches: The left image shows how an RGB camera reveals identifiable information, while the right image demonstrates the anonymized depth data provided by a ToF camera.

ToF cameras are strategically installed above doorways to monitor occupant entry (IN) and exit (OUT) movements. This overhead placement provides an unobstructed view of the passageway, reduces the likelihood of occlusions caused by furniture or overlapping



FIGURE 1 Comparison between RGB image and Time-of-Flight (ToF) depth data.

individuals, and minimizes the number of cameras required for comprehensive monitoring. By analyzing changes in depth patterns, the system detects directional movements and dynamically updates the occupancy count in real time. This configuration focuses solely on doorway activity, simplifying monitoring and enabling precise control of energy-consuming systems based on real-time occupancy data. Figure 2 illustrates the placement and function of the ToF camera in detecting IN and OUT movements.

2.2 | Mean shift clustering for occupant detection

To accurately detect and track occupants in the ToF camera's depth data, the proposed system employs the mean shift clustering algorithm [24]. The entire process, from data point construction to clustering, is depicted in Figure 3. The preprocessing steps ensure that the data focus exclusively on occupants while eliminating noise and irrelevant information. First, the image is processed to remove static obstacles, leaving only the moving occupants. This can be achieved either by subtracting a reference image containing only obstacles or by cropping the image to include only the pathway where occupants pass through. In this study, the latter approach is adopted, where the region of interest (ROI) is defined to capture only the passageway. Additionally, since high-resolution images are unnecessary for this application, the image is compressed to an appropriate level to improve computational efficiency. Figure 3A,B demonstrates the processes of image compression and cropping to isolate the ROI.

The ToF camera outputs depth images, where depth information is represented by color gradients. For instance, closer objects are rendered in shades of red, while farther objects are displayed in shades of blue. The

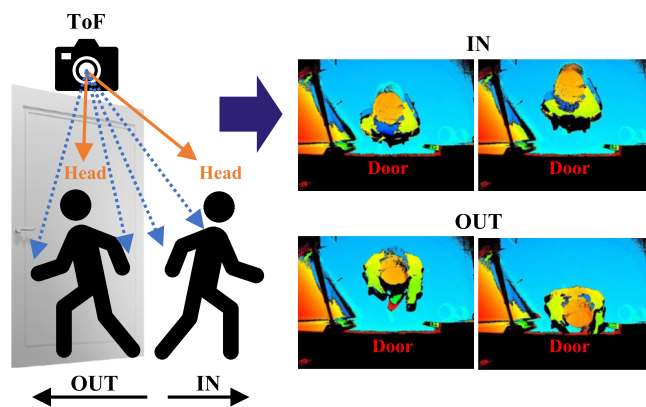


FIGURE 2 Illustration of the Time-of-Flight (ToF) camera placement and functionality.

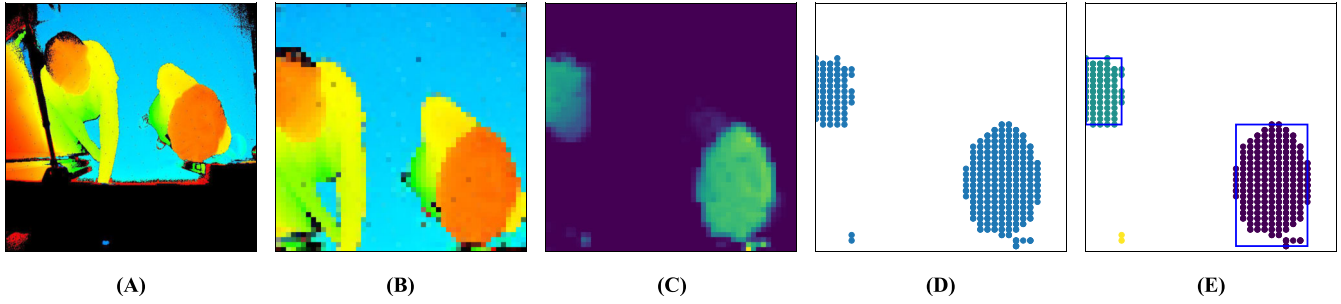


FIGURE 3 Overall process of clustering-based object detection: (A) Original image, (B) compression and crop, (C) extracting R component, (D) generating data points, and (E) clustering.

specific color mapping can be customized by the user, but it is essential that closer and farther objects are distinguishable. Typically, the head of an occupant is represented in a color closer to red, which corresponds to a high R (red) component in the RGB representation of the depth image. To focus on these high R component regions, the pixel intensity values are transformed using a weighted sum of the RGB components. Let $R(i)_{j,k}$, $G(i)_{j,k}$, and $B(i)_{j,k}$ represent the red, green, and blue values of the pixel at position (j, k) in the i -th image. The transformed pixel intensity $I(i)_{j,k}$ is defined as follows:

$$I(i)_{j,k} = \max(w_1 R(i)_{j,k} + w_2 G(i)_{j,k} + w_3 B(i)_{j,k}, 0), \quad (1)$$

where w_1 , w_2 , and w_3 are weights assigned to the R , G , and B components, respectively. For example, setting $w_1 = 1$, $w_2 = -1$, and $w_3 = -1$ isolates regions with a high $R(i)_{j,k}$ component. Consequently, pixels with higher $R(i)_{j,k}$ values will have larger $I(i)_{j,k}$ values, while pixels with lower $R(i)_{j,k}$ values will be suppressed to near-zero. Figure 3C highlights how the relevant regions are extracted.

To represent the spatial distribution of high-intensity regions, a multiset $X(i)$ is constructed for the i -th image. This multiset maps pixel intensity values to the number of spatial data points. The multiplicity $m_{X(i)}((j,k))$ represents the number of data points at position (j,k) generated from that pixel intensity $I(i)_{j,k}$ and is defined as follows:

$$m_{X(i)}((j,k)) = \left\lfloor \frac{I(i)_{j,k}}{s} \right\rfloor, \quad (2)$$

where s is a scaling factor that prevents overly high $I(i)_{j,k}$ values from producing an excessive number of data points. This ensures that the dataset remains computationally manageable. As a result, regions with high $R(i)_{j,k}$ components, corresponding to occupant heads, are

effectively emphasized, creating a concentrated set of data points in $X(i)$ for further processing. Figure 3D illustrates how pixel intensity values are transformed into data points, concentrating on high-intensity regions.

Once the data points are constructed, clustering is performed to group them into clusters. The number of clusters detected corresponds to the number of occupants in the image. Therefore, clustering algorithms that require a predefined number of clusters, such as k -means clustering, are unsuitable for this study. Instead, a clustering algorithm that can adaptively determine the number of clusters is necessary. We employ the mean shift clustering algorithm, as it is well-suited for this task. The algorithm assumes that the data points form Gaussian distributions centered around occupant heads, allowing for robust clustering without prior knowledge of the number of occupants. Also, during the data point construction phase, it is possible for data points to form in areas other than occupant heads. However, clustering algorithms inherently provide anomaly detection capabilities [25]. Clusters with insufficient data points are automatically discarded as noise, ensuring that only valid occupant-related clusters are retained. The clustering results are depicted in Figure 3E, where each cluster represents an occupant detected in the scene.

The mean shift clustering algorithm is a nonparametric method that identifies clusters by iteratively shifting data points towards areas of high density [24]. This approach is particularly suitable for the proposed system as it does not require prior knowledge of the number of clusters, adapting dynamically to the spatial distribution of data points. The algorithm begins with kernel density estimation (KDE), which estimates the probability density function (PDF) of data points. With the set of data points $X(i)$, the density at a point (j', k') is estimated using a kernel function $K(\cdot)$ as follows:

$$f((j', k')) = \frac{1}{nh^2} \sum_{(j,k) \in X(i)} K\left(\frac{\sqrt{(j'-j)^2 + (k'-k)^2}}{h}\right), \quad (3)$$

where n is the number of data points in $X(i)$ and h is the bandwidth parameter that controls the kernel's smoothing range. Typically, $K(\cdot)$ is a Gaussian function defined as follows:

$$K(u) = \exp\left(-\frac{u^2}{2}\right). \quad (4)$$

The KDE process is visualized in Figure 4, where the density of one-dimensional data is estimated. Peaks in the density function represent high-density regions, which correspond to cluster centers in higher-dimensional data. These cluster centers form the basis for accurately detecting occupants in our system, as they align with the spatial distribution of occupant-related data points. This provides the foundation for identifying meaningful groups of data points in the subsequent clustering process.

Using this density estimation, the algorithm computes the cluster center, which represents the direction and magnitude of movement towards the highest density region. The cluster center for a point $(\hat{j}, \hat{k})$ is defined as follows:

$$\Delta((\hat{j}, \hat{k})) = \frac{\sum_{(j,k) \in X(i)} (j - \hat{j}, k - \hat{k}) K\left(\frac{\sqrt{(j - \hat{j})^2 + (k - \hat{k})^2}}{h}\right)}{\sum_{(j,k) \in X(i)} K\left(\frac{\sqrt{(j - \hat{j})^2 + (k - \hat{k})^2}}{h}\right)}. \quad (5)$$

Initially, $(\hat{j}, \hat{k})$ is set to one of the points $(j, k) \in X(i)$. Then, the algorithm iteratively updates the position of $(\hat{j}, \hat{k})$ using the following update rule:

$$(\hat{j}, \hat{k}) \leftarrow (\hat{j}, \hat{k}) + \Delta((\hat{j}, \hat{k})). \quad (6)$$

The iterative process continues until $(\hat{j}, \hat{k})$ converges to a high-density peak, where the movement $\Delta((\hat{j}, \hat{k}))$

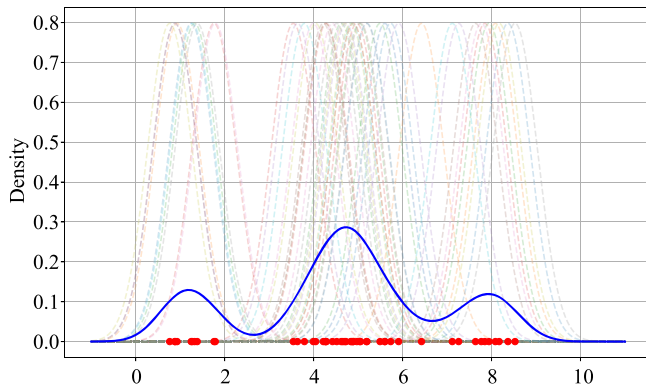


FIGURE 4 Kernel density estimation (KDE) visualization.

becomes negligible. This procedure is repeated for all points in $X(i)$. After convergence, points that are within a certain radius of the same cluster center are grouped into the same cluster. Clusters with insufficient data points are discarded as noise to ensure robustness against anomalies. The final cluster centers represent the detected occupants in the image, and the total number of clusters corresponds to the number of occupants detected.

The iterative process is illustrated in Figure 5, where the KDE contours represent the density function in two dimensions, and data points are shown shifting towards the peaks of the density. As data points converge to the highest density regions, clusters naturally form. This step-by-step process effectively identifies clusters without requiring prior knowledge of their number. The KDE and mean shift processes, as depicted in Figures 4 and 5, demonstrate how density estimation and iterative shifting are used to identify clusters corresponding to occupant heads.

2.3 | Occupancy tracking and counting

Once occupants are detected in the image using clustering, the next step is to determine whether each detected occupant is entering (IN) or exiting (OUT) the monitored space. While advanced tracking algorithms like SORT (simple online real-time tracking) could be used, they are less effective in scenarios where bounding box shapes change significantly, such as when occupants pass through doorways [26]. This limitation arises because SORT primarily tracks objects based on bounding box consistency, which is disrupted in such cases. To address this, we propose a simplified tracking algorithm designed specifically for occupancy counting.

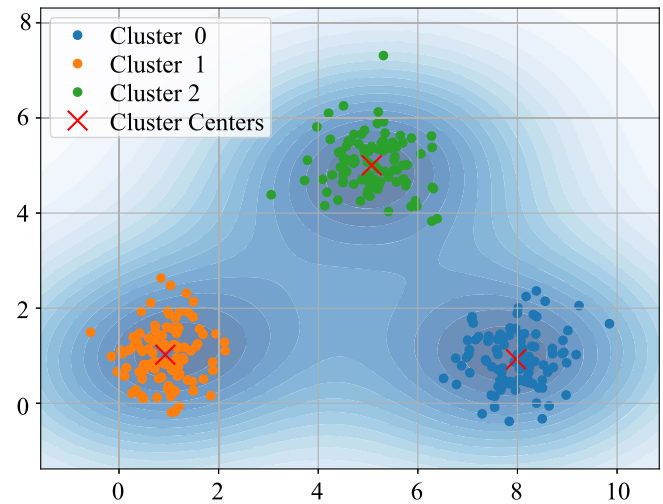


FIGURE 5 Mean shift clustering with kernel density estimation (KDE) contours.

To facilitate tracking and counting, bounding boxes are generated based on the clustering results. For each cluster c in image i , the minimum and maximum j -values and k -values of all data points in the cluster are determined. These values, denoted as $j_{i,c,\min}$, $j_{i,c,\max}$, $k_{i,c,\min}$, and $k_{i,c,\max}$, define a rectangular bounding box $B(i,c)$ for cluster c in image i :

$$B(i, c) = [j_{i,c,\min}, j_{i,c,\max}, k_{i,c,\min}, k_{i,c,\max}]. \quad (7)$$

Given that the ToF camera is positioned above a doorway, one edge of the image corresponds to the doorway. This study assumes that the bottom edge of the image represents the doorway. To track clusters across consecutive frames, certain conditions must be evaluated for cluster c in image i and cluster $\tilde{c}$ in the subsequent image $i+1$. Since we are only interested in bounding boxes near the doorway, we extract only those clusters c and $\tilde{c}$ that satisfy the following conditions.

$$k_{i,c,\min} < \alpha, k_{i+1,\tilde{c},\min} < \alpha, \quad (8)$$

where α determines whether the bounding box is near the doorway. When an occupant enters or exits the monitored space, the bounding box near the doorway will exhibit minimal horizontal variation. To track clusters across consecutive frames, the following conditions are evaluated:

$$|j_{i+1,\tilde{c},\max} - j_{i,c,\max}| < \tau, |j_{i+1,\tilde{c},\min} - j_{i,c,\min}| < \tau, \quad (9)$$

where τ ensures that the bounding box width remains consistent across frames. If these conditions are satisfied, clusters c in image i and $\tilde{c}$ in subsequent image $i+1$ are determined to represent the same occupant, enabling reliable tracking across frames. The final determination of whether the occupant is entering (IN) or exiting (OUT) is based on the change in the vertical length of the bounding box between frames. This is computed as follows:

$$\text{Determination} = \begin{cases} \text{IN}, & \text{if } \Delta k > 0, \\ \text{OUT}, & \text{if } \Delta k < 0. \end{cases} \quad (10)$$

Here, Δk is defined as the change in the vertical length of the bounding box between frames:

$$\Delta k = (k_{i+1,\tilde{c},\max} - k_{i+1,\tilde{c},\min}) - (k_{i,c,\max} - k_{i,c,\min}). \quad (11)$$

The assumption that the camera's frame rate is sufficiently high is crucial for this step. A high frame rate ensures that the same occupant is captured in consecutive

frames without significant positional shifts or distortion in the bounding box shape. This assumption is reasonable for modern ToF cameras, which are designed to operate at high frame rates, providing smooth and reliable data for tracking. Furthermore, since the monitoring area is limited to the doorway, rapid changes in occupant positions are unlikely, supporting the validity of this assumption. If these conditions are satisfied, cluster c in image i and cluster $\tilde{c}$ in image $i+1$ are determined to represent the same occupant. Based on experimental validation, it was found that using 10 consecutive frames was optimal for accurately tracking occupants entering or exiting. If the ToF camera had a lower frame rate or experienced dropped frames, it would be unable to detect the small positional changes in the bounding box near the doorway, leading to potential failures in occupant counting. While a high frame rate results in many captured frames, there is no significant overhead in this approach, as images that have already been used for occupant counting are deleted, ensuring that the system remains efficient and does not accumulate unnecessary data.

Algorithm 1 Occupancy IN/OUT Determination

```

1: Input: Sequence of ToF images
2: Output: IN/OUT status of detected occupants
3: for each image do
4:   Perform preprocessing on image  $i$ 
5:   Extract data points  $X(i)$  from depth information
6:   Apply clustering algorithm (e.g., mean shift) to  $X(i)$ 
7:   Generate bounding boxes  $B(i, c)$  for each cluster  $c$ 
8: end for
9: for each pair of image  $i$  and image  $(i+1)$  do
10:   for each cluster  $c$  in image  $i$  and  $\tilde{c}$  in image  $(i+1)$  do
11:     if satisfying (8 and (9)) then
12:       Match  $c$  with  $\tilde{c}$ 
13:       Compute  $\Delta k$  from (11)
14:       if  $\Delta k > 0$  then
15:         Label: IN
16:       else if  $\Delta k < 0$  then
17:         Label: OUT
18:       end if
19:     end if
20:   end for
21: end for

```

The process is illustrated in Figure 6, which shows the bounding boxes and their transitions between frames, highlighting how vertical length changes are used to determine IN/OUT movements. This approach ensures robust tracking and accurate classification of IN and

OUT movements based on bounding box dynamics, without relying on complex appearance-based tracking algorithms. The simplicity of this method makes it computationally efficient and well-suited for real-time occupancy detection. The step-by-step process for detecting occupants and determining their IN/OUT status is outlined in Algorithm 1 and visually summarized in Figure 7. The flowchart in Figure 7 also serves to represent the overall system architecture, which consists of modular processing stages: preprocessing, clustering, tracking, and final classification. This pipeline-based design supports interpretability, lightweight implementation, and labeling-free operation.

3 | EXPERIMENTAL RESULTS

3.1 | Dataset and evaluation scenarios

The experimental setup involved positioning a ToF camera above a doorway to monitor occupant movements. The camera was carefully placed to ensure that the motion of the door itself did not interfere with the

captured depth data. The dataset used for evaluation consists of approximately 270 depth frames, with an average of 10 frames captured per event, collected from 27 distinct entry/exit events in a real indoor setting. The test site involved a narrow doorway leading to a single room, which naturally constrained most movement to single-person entries or exits. This distribution reflects the intended application of the proposed system: monitoring room-level occupancy in smart buildings for zone-specific energy optimization, such as targeted heating, ventilation, and air conditioning (HVAC) control. In this context, the goal is not to track large crowd movements at building entrances but to detect the presence of individuals entering or exiting specific rooms.

Consistent with this goal, the physical environments in which such systems are deployed—offices, dormitories, meeting rooms, and similar enclosed spaces—are typically designed to accommodate single-person passage. According to general architectural design standards, interior doors in such spaces are commonly constructed with a clear width of approximately 800 mm to 900 mm, which comfortably allows one person at a time to pass through. This design standard is widely adopted due to its efficiency in space usage, cost-effectiveness, and alignment with anthropometric norms [27]. As such, individual entry and exit are the dominant patterns observed in real-world settings of this type.

Table 1 summarizes the composition of single-person occupancy scenarios included in the dataset. Due to the spatial limitations of the test environment and safety considerations with only a few participants, complex scenarios involving multiple persons passing through simultaneously were naturally limited during data

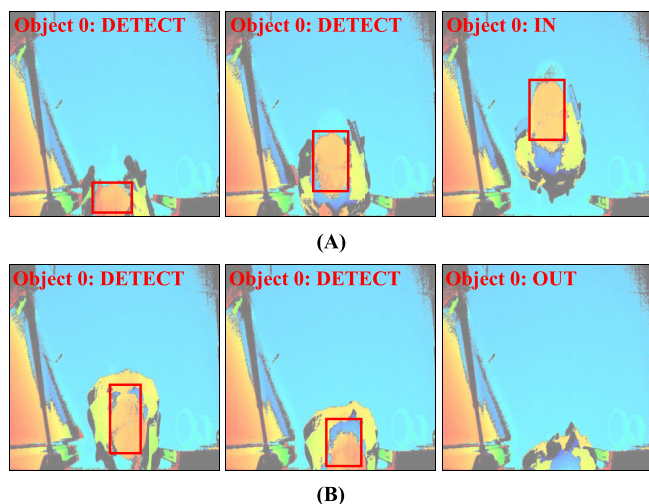


FIGURE 6 Occupant tracking process: (A) Example of tracking a single occupant entering (IN) and (B) example of tracking a single occupant exiting (OUT).

TABLE 1 Composition of occupancy scenarios.

Scenario type	Number of events	Number of frames
Single entry (IN)	13	~130
Single exit (OUT)	14	~140
Total	27	~270

Note: The bolded row for “Total” emphasizes the sum of all events and frames, providing a quick overview of the dataset size.

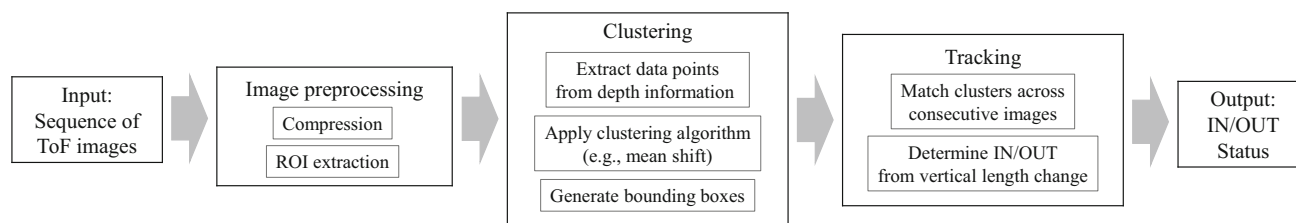


FIGURE 7 Overall flowchart of the proposed labeling-free occupancy counting algorithm.

collection. However, to ensure the robustness of the proposed method across less frequent yet practically relevant cases, we deliberately included at least one representative instance of multiple entry, multiple exit, and simultaneous bidirectional movement. Although these cases are rare in the intended deployment setting, their inclusion serves to validate the algorithm's behavior under edge-case dynamics.

Since the proposed method is based on unsupervised clustering and does not involve any supervised training process, the conventional notion of train/test data splits does not apply. Instead, a small subset of initial frames was used to manually tune hyperparameters in Table 2. The entire dataset was then used collectively for evaluating the system's performance across representative occupancy scenarios. To determine appropriate hyperparameter values, we selected one representative single-entry and one single-exit scenario and performed parameter sensitivity tests by systematically varying the values and observing the resulting detection behavior. Parameters such as the bandwidth h and the thresholds α and τ were tuned based on this limited subset. Given the unsupervised nature of the method and the simplicity of the parameter space, this approach is consistent with practical deployment, where labeled data are unavailable and exhaustive tuning is infeasible.

Although only a single pair of scenarios was used for tuning, the selected parameters generalized well across all remaining scenarios. This is supported by the high detection accuracy observed during evaluation, indicating that the method is not highly sensitive to minor parameter changes and that limited tuning is sufficient for stable, real-world performance. This limited tuning strategy also aligns with real-world deployment practices, where system installers can easily calibrate parameters using a single representative scenario from the target environment without requiring labeled data or extensive testing.

TABLE 2 List of hyperparameters and their values.

Hyperparameters	Values
Compressed image size	(67, 48)
Weight assigned to the R component (w_1)	1
Weight assigned to the G component (w_2)	-1
Weight assigned to the B component (w_3)	-1
Scaling factor for generating the data points (s)	50
Bandwidth in mean shift clustering algorithm (h)	6
The threshold for determining proximity to the doorway (α)	13
The tolerance for bounding box consistency (τ)	4

3.2 | Results and analysis

A common question raised when the proposed method is why pretrained YOLO models cannot be used directly for occupancy detection. To address this, we compared the performance of the proposed method with a pretrained YOLO model without fine-tuning. Figure 8A shows the results of occupancy detection using the proposed approach, while Figure 8B illustrates the detection results of a pretrained YOLO model applied to the same scene.

In Figure 8B, the pretrained YOLO model failed to detect the occupant on the left entirely, while the occupant on the right was detected redundantly with multiple bounding boxes, and the detection results were incorrectly classified as non-human objects. These shortcomings highlight the fundamental limitation of YOLO when applied to ToF images. YOLO is designed for identifiable RGB images and relies heavily on detailed visual features for accurate detection. Consequently, it is not suited for privacy-preserving ToF images, where identifiable features are intentionally absent.

Moreover, these results demonstrate that fine-tuning YOLO for ToF images is essential to achieve any level of acceptable performance. However, fine-tuning YOLO for this purpose would require an extensive dataset of labeled ToF images, which is not only resource-intensive to create but also inefficient compared to the proposed approach. By contrast, the proposed method inherently avoids these challenges by leveraging clustering-based detection on depth data, which requires no labeled datasets and preserves privacy. This comparison underscores the practicality and efficiency of the proposed method over YOLO for occupancy detection in privacy-sensitive environments.

Although various image detection and classification techniques have been proposed in the literature, YOLO was chosen as the sole baseline for comparison due to its status as a widely recognized state-of-the-art model for real-time object detection. YOLO is also commonly applied in occupancy estimation tasks in recent smart

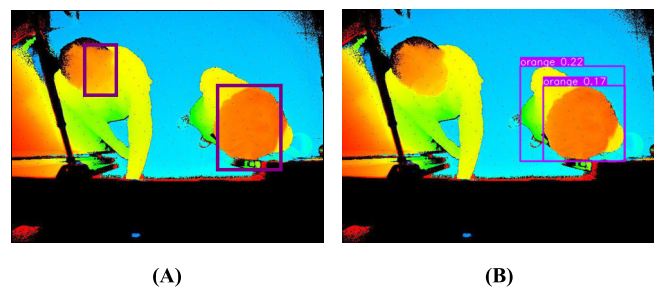


FIGURE 8 Comparison between the proposed method and pretrained you only look once (YOLO): (A) Result of the proposed method and (B) Result of the pretrained YOLO.

building and surveillance applications [6]. Given its prominence and real-time capability, we deemed it the most appropriate supervised benchmark for comparison. Other methods were therefore not included to maintain focus and clarity in evaluation.

To evaluate the performance of the proposed method, we conducted a comparative analysis against pretrained YOLO [6], k -means clustering [22], and DBSCAN [28] on the same dataset. To the best of our understanding, there are few, if any, existing approaches that apply unsupervised clustering to ToF depth imagery for labeling-free occupancy detection. Therefore, we focused our comparative analysis on alternative clustering methods—namely, k -means clustering and DBSCAN—which align with the unsupervised, image-based nature of our proposed method. The results are summarized in Table 3. Although YOLO could potentially outperform unsupervised methods if fine-tuned with labeled ToF data, such an approach would violate the labeling-free constraint central to this study. For fairness and consistency, we used the pretrained YOLO model without any fine-tuning. While YOLO was able to form bounding boxes to some extent—providing partially trackable detections—it often failed to distinguish individual occupants or to form complete bounding boxes. These limitations led to significantly lower accuracy values despite its relatively long execution time.

For all clustering-based methods, hyperparameters were determined using the same reference scenario used for configuring the proposed mean shift method. In the case of k -means, since the number of clusters k must be specified in advance, we applied silhouette score analysis for multiple k values on a per-image basis to identify the optimal cluster count. However, due to its sensitivity to noise, k -means exhibited reduced performance on ToF images with sensor distortions. DBSCAN, on the other hand, showed more robust clustering behavior under noisy conditions and achieved strong performance comparable to mean shift. Nevertheless, mean shift consistently demonstrated the highest accuracy across both single-entry and single-exit cases. While this suggests its relative superiority in our specific test environment, we acknowledge that performance may vary depending on

deployment scenarios. The main contribution of this work lies not in the specific choice of clustering algorithm but in the overall framework enabling occupancy detection under labeling-free constraints.

The proposed mean shift algorithm was evaluated under conditions with environmental noise and lighting variations, which are common challenges in ToF imaging. While no method can achieve perfect accuracy in such conditions, mean shift demonstrated robustness against noise, performing better than k -means clustering. k -means is highly sensitive to outliers, which causes its performance to degrade significantly when noise is present. In contrast, mean shift's density-based approach allows it to handle noise more effectively, leading to more reliable occupant detection even in the presence of environmental fluctuations. However, despite this advantage, the system's accuracy did not reach 100%, highlighting the ongoing challenge of noise in ToF data.

In terms of execution time, pretrained YOLO required the most processing time per event due to its complex architecture. Among the clustering-based approaches, execution times were similar, with DBSCAN being the fastest and mean shift the slowest. Although mean shift involved slightly longer processing time than the others, the difference was marginal. Notably, k -means required additional time due to the per-image evaluation of multiple cluster counts for silhouette score optimization, which offset its simplicity in clustering logic.

To complement the quantitative results, Figure 9 provides qualitative examples of how the proposed method operates in single-person scenarios. Specifically, Figure 9A shows a case of single entry, and Figure 9B shows a case of single exit. In both examples, the system successfully detects and tracks the occupant's movement, leading to correct IN/OUT classification. These visualizations support the effectiveness of the proposed pipeline as demonstrated by the numerical evaluation.

To evaluate the robustness of the proposed method beyond typical single-person scenarios, we included three complex test cases: multiple entry, multiple exit, and simultaneous bidirectional movement. Although the physical doorway was designed to accommodate single-person passage—as is common in real-world

TABLE 3 Performance comparison across methods on single-person entry/exit scenarios.

Method	Entry accuracy	Exit accuracy	Avg. time per event (s, CPU)
Pretrained YOLO	23.1%	35.7%	3.57
k -means clustering	84.6%	85.7%	1.41
DBSCAN	92.3%	85.7%	1.23
Mean shift	92.3%	92.9%	1.55

Note: The bolded values for the mean shift method are the best performance results (highest accuracy) achieved in the comparison.

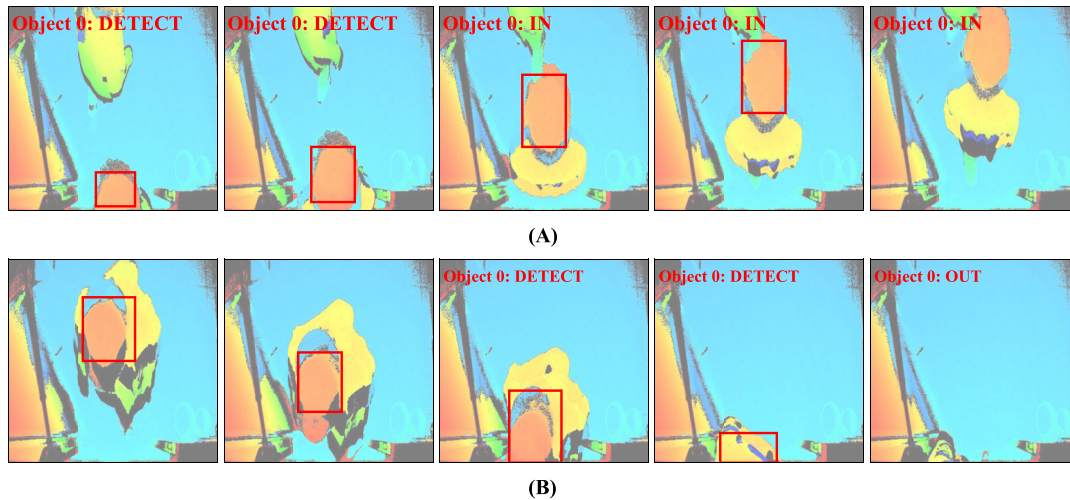


FIGURE 9 Experimental results in single-person entry/exit scenarios: (A) single occupant entering and (B) single occupant exiting.

TABLE 4 Success/failure comparison on complex scenarios.

Method	Multiple entry	Multiple exit	Simultaneous entry and exit
Pretrained YOLO	Failure	Failure	Failure
<i>k</i> -means clustering	Failure	Failure	Failure
DBSCAN	Success	Success	Success
Mean shift	Success	Success	Success

indoor environments—these edge-case scenarios were deliberately introduced to assess the system's behavior under more challenging conditions. Each scenario involved exactly two individuals and occurred only once due to spatial and safety constraints, which made conventional accuracy metrics inapplicable. Instead, we evaluated each method based on binary success or failure in detecting and classifying the occupants correctly.

As summarized in Table 4, the pretrained YOLO model failed in all three scenarios. It was unable to form appropriate bounding boxes when multiple people appeared in the frame, often merging them into a single detection or missing them entirely. *k*-means, despite using silhouette score optimization to select the number of clusters on a per-image basis, also failed to distinguish closely positioned individuals, frequently collapsing them into one cluster. By contrast, both DBSCAN and the proposed mean shift approach successfully handled all three scenarios. These methods exhibited robustness to sensor noise inherent in ToF images and effectively separated individuals without requiring supervision. Among them, mean shift demonstrated superior practical characteristics: It did not require per-image parameter tuning, produced smoother clustering boundaries, and supported more stable tracking across frames. Therefore,

although DBSCAN also performed well, mean shift is recommended for its consistency, usability, and alignment with the labeling-free objective of this study.

Figure 10 illustrates the successful performance of the proposed method in these complex cases. Specifically, Figure 10A shows accurate tracking in the multiple entry scenario, Figure 10B in the multiple exit scenario, and Figure 10C in the simultaneous bidirectional movement case. These visual examples reinforce the system's ability to manage overlapping movements and multiple occupants even under challenging real-world conditions.

The proposed mean shift algorithm does not have a strict upper limit on the number of clusters it can detect; however, its performance may degrade as the number of occupants increases due to overlapping clusters or diminishing density. In practice, the number of clusters corresponds to the number of distinct individuals, and as the number of people in a frame increases, the algorithm may require additional tuning to accurately separate them. However, it is important to note that the system was specifically designed for occupancy detection in indoor environments such as offices or classrooms, where typically only one or two individuals are present in the doorway at any given time. Given the nature of the monitored space, it is unlikely that the system will need to

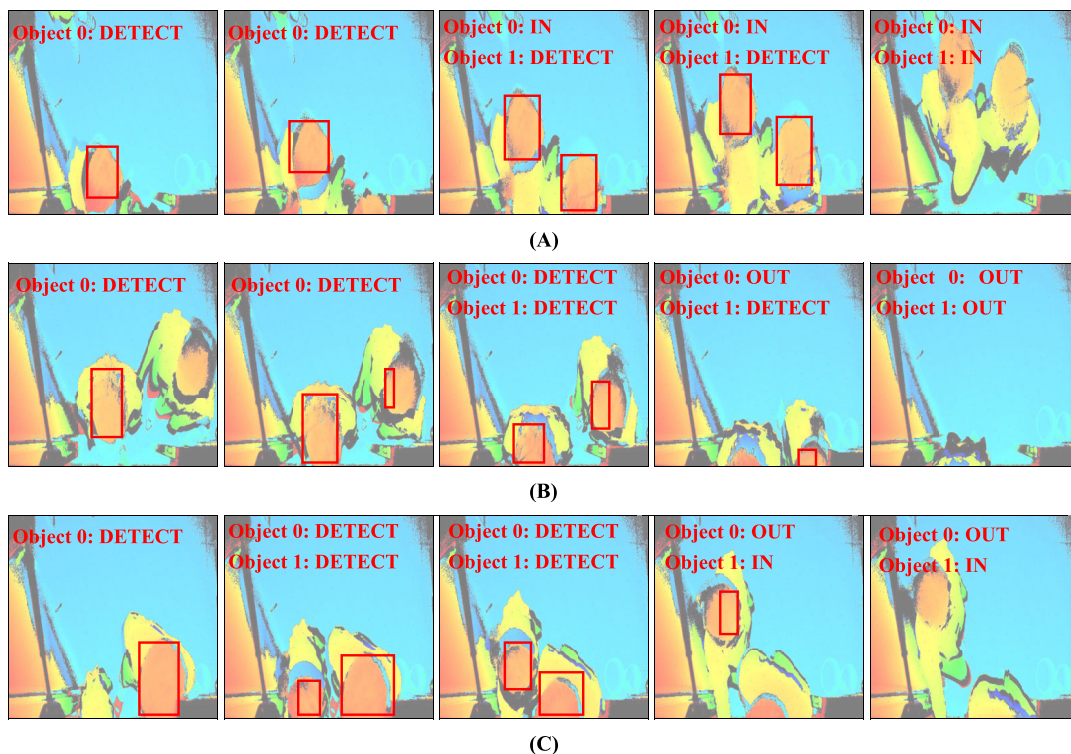


FIGURE 10 Experimental results in complex scenarios: (A) multiple occupants entering simultaneously, (B) multiple occupants exiting simultaneously, and (C) simultaneous entry and exit events.

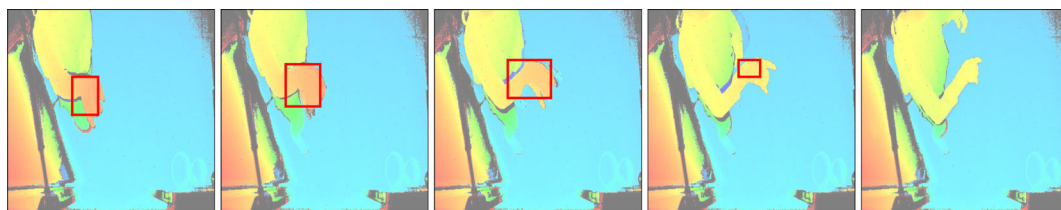


FIGURE 11 Example of anomalous cases where a raised hand is erroneously identified as an occupant.

handle large numbers of occupants simultaneously, thus making the current approach well-suited to the problem scope.

3.3 | Limitations and discussions

While the proposed occupancy detection system demonstrates strong performance in standard entry and exit scenarios, certain inherent assumptions in the methodology can lead to errors in more complex situations. A key assumption is that the highest point of an object passing through the monitored area corresponds to the head of an occupant. While this assumption holds true in most cases, it is not universally applicable. For example, if an individual raises their hand, the system may misclassify the hand as a separate occupant. Similarly, in cases

where two individuals, such as a couple or a parent with a child, enter or exit closely together, the system may detect them as a single occupant. Nonhuman objects, such as furniture or packages, passing through the doorway can also be misclassified as occupants. Figure 11 illustrates a scenario where a raised hand is erroneously identified as an occupant. This issue arises when the hand is positioned above the head, causing the system to misclassify it as an additional individual. A similar problem can occur when an occupant waves their hand near the entrance, leading to an erroneous count of one extra occupant. This limitation is primarily caused by the system's reliance on the highest point of the object passing through the doorway, which is not always representative of a human head.

Additionally, when occupants move at extremely high speeds through the monitored area, the system may fail

to properly detect them in consecutive frames. This issue arises from the reliance of the proposed algorithm on α and τ values to determine IN and OUT statuses. Although this study assumes that the camera's frame rate is sufficiently high to capture occupants' movements accurately, errors can occur if the camera's frame rate is unable to keep up with the speed of movement. In such cases, the proposed method may fail to track occupants effectively, leading to inaccuracies in counting.

We acknowledge that these limitations are not unique to the proposed method but are inherent to any system relying on single-sensor modalities [29]. Even YOLO would face significant challenges in such scenarios. For example, if an occupant moves too quickly for the camera's frame rate, YOLO would also struggle to maintain accurate tracking or detection, particularly in environments where privacy-preserving ToF images lack detailed visual features. While YOLO-based approaches might seem like a solution to other anomalies, they require extensive labeled datasets covering scenarios such as raised hands, closely entering individuals, or objects passing through the monitored area. These datasets must be recollected and relabeled for each new environment, making this approach resource-intensive and impractical. Moreover, YOLO would need high-resolution ToF image data and computationally expensive models to achieve similar detection accuracy, which are unsuitable for low-power devices like Raspberry Pi and costly for deployment across multiple entry points.

To address the identified limitations, integrating complementary sensing modalities, such as passive infrared (PIR) sensors, acoustic sensors, or wireless channel state information (CSI) signals, can significantly enhance the robustness and accuracy of the system [30]. PIR sensors, for instance, can confirm occupancy presence in scenarios where the ToF camera may misclassify nonhuman objects or anomalies, such as raised hands or furniture. Acoustic sensors can detect human activity or footsteps, serving as a secondary verification mechanism during high-speed movements or overlapping entry and exit scenarios. CSI-based techniques, which analyze variations in wireless signal propagation, may offer an additional privacy-preserving layer by detecting motion or presence through wall-penetrating signals without relying on image data. These multimodal approaches not only improve the system's reliability in complex situations but also preserve its lightweight and privacy-focused design. The composite sensor system that combines additional sensor modalities with the current method offers a robust framework for accurate and privacy-conscious occupancy detection [31].

Beyond the technical limitations, the scalability and deployment assumptions of the system must also be

discussed. The proposed framework is specifically designed for small-scale, room-level occupancy detection, where individuals pass through narrow doorways one at a time. This assumption is valid in common real-world environments such as offices and classrooms, where the doorway width inherently limits crowd flows. In such settings, multiple doorways can be handled independently by deploying a ToF camera at each location. The system does not support large-scale crowd tracking or overlapping movements in open areas (e.g., train stations or stadiums), where multicamera fusion or predictive modeling would be required. Additionally, deployment assumes that the camera is installed such that door motion is not captured within the frame—positioned on the side opposite to the door's swing direction—to avoid occlusion and motion artifacts. These assumptions, while practical and easily implementable, also delineate the current system's applicable scope. Future work may explore adaptations for larger-scale environments and integration with other modalities to enhance generalizability.

Although the evaluation was conducted in a controlled environment, the system is designed with minimal hardware and parameter dependencies, making it readily deployable for long-term use in real-world indoor settings such as offices or classrooms. Its labeling-free operation and lightweight processing requirements allow stable operation over extended periods without manual intervention or recalibration. While the detection accuracy does not reach 100% and errors may accumulate in long-term deployment, such errors can be partially mitigated by integrating a composite sensor framework that includes PIR or acoustic sensors. These complementary sensors, while not capable of occupant counting, can reliably detect presence or absence. When no presence is detected, the system can safely reset the occupancy count to zero, helping to bound cumulative error and enhance robustness in practical applications.

4 | CONCLUSIONS

This study proposed a privacy-preserving and labeling-free occupancy detection system leveraging ToF cameras and a clustering-based approach. By eliminating the reliance on large labeled datasets and identifiable image data, the proposed method addresses critical challenges in privacy-sensitive environments while ensuring computational efficiency and scalability. Experimental results demonstrated that the proposed system achieves over 90% detection accuracy in standard indoor scenarios, despite the presence of ToF sensor noise. The method also successfully handled more complex situations such as multiple entry, multiple exit, and simultaneous

bidirectional movement. These results highlight the system's robustness and practical applicability for privacy-preserving, real-time occupancy monitoring without the need for labeled data.

While limitations such as handling anomalous behaviors and high-speed movements were identified, these challenges are not unique to the proposed method and present opportunities for future work. Future research will focus on improving the system's ability to handle complex scenarios, such as unconventional occupant positions or non-human objects, by integrating additional sensors like sound or PIR and developing advanced algorithms. Furthermore, to enhance its robustness in dynamic environments, future work will explore adaptive parameter mechanisms that can automatically tune the system based on varying conditions or crowd densities. Additionally, future extensions may include adapting the system to large-scale, multidoorway environments through multicamera integration, and scalable coordination strategies.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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